SAM Based Masking for PIV Experiments

MIE498Y: Research Thesis

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Date

Abstract

The abstract is generally 100 to 200 words, single-spaced in block form, placed in the centre of a separate page.

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B	Example definition

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1 Introduction

1.1 Particle Image Velocimetry (PIV)

Particle Imagery Velocimetry (PIV) is a flow visualization technique that measures the instantaneous velocity of the fluid flow by calculating the displacement of laser illuminated particles in two different images taken, this is also known as the an image pair in the context of PIV experiments. In PIV experiments, small particles are seeded into the fluid flow, and photos will be taken for those particles as they became illuminated by laser. By comparing the displacement of the particles seen from the image pair and assuming that the motion of particles follows the motion of the fluid, the fluid velocity field can be calculated.

1.2 PIV Masking

The use of laser in the experiment resulted in measurement errors near object boundaries due to the reflection from the object. To resolve this issue, masks are created to exclude certain areas from the solution algorithm. Traditionally, manual masking is performed by researchers to define zones of exclusion. However, such method is impractical for measurement of moving objects as the experiment usually uses high speed camera and has a significant amount of data needing labeling. Some automated masking procedures are developed to help with these scenario. One of them involves fitting predetermined polynomial to the PIV data as the output mask. However this method requires preexisting knowledge about the motion and shape of the object measured, which would be impractical in some cases.

1.3 Computer Vision

Computer vision is a sub-field of artificial intelligence which focused on enabling machines and algorithms to interpretate visual data, usually in the form of images or videos. The origin of computer vision can be traced back to the 1960s. Today computer vision is used for a range of tasks, from image denoising to segmentation and object labeling.

1.4 Computer Vision for PIV

Computer vision has been proposed as a measure for mask generation in PIV experiments. The proposed algorithm uses the Segment Anything Model developed by Meta AI capable of zero-shot generalization, to create masks around bounding points determined by the Grounding-Dino model, capable of identifying location of objects using natural language input. This inherently simplifies the workflow of the masking procedure and provides alternative measure to masking objects in motion in PIV experiments. This paper analyzes the aforementioned computer vision algorithm used in PIV experiment by adapting it to a python based PIV solver. The output from the modified solver will then be validated against manual masking and polynomial fitting to obtain the final conclusion regarding the feasibility of using computer vision for PIV experiments.

2 Literature Review

This chapter introduces coding libraries and literature critical to the research discussed in this report.

2.1 OpenPIV

OpenPIV is an open-source, community driven project on the development of a free PIV analysis and post-processing software. The python version of the package, OpenPIV-Python is currently available throught the Python Package Index (PyPI). This package offered flexible and easy to use interface for the processing of PIV data in the form of image pairs. The package is available both in the code format and in the GUI format for both windows and linux platforms.

The OpenPIV library provides access to correlation algorithm essential to the computation of velocity fields, and validation through calculation which is studied in this report. Due to the noisy nature of the immediate object boundary of objects embedded in flow as a result of reflection. The signal to noise ratio validation is of specific interest to the subject of study in this report. This library provides the method to analyze the efficiency of the masks identified by the computer vision algorithms by investigating the effect if has on removing significantly noisy layers immediate to object boundaries.

2.2 OpenPIV-Python-CPU

OpenPIV-Python-CPU is a library developed from the code of OpenPIV-Python and optimized for the perfomrance when the software is ran on the Central Processing Unit (CPU). While similar to the OpenPIV-Pyhton code from many aspects, this library uses different correlation algorithm and interogation windows sizing in hopes of improved performance. The database uses a different but similar syntax for its operation compared to the OpenPIV-Python module. While providing simplifications to the model, this model is shown to construct the fluid velocity profile to a satisfactory degree. This model also provides signal to noise validation that is critical for the validation step of this research.

This module will be used as the main algorithm for the computation of PIV data for the aforementioned reasons and the availability of tutorials in the form of jupyter notebooks. The code used for this research are modified on the basis of those tutorials as they provide a proven workflow for PIV analysis. The mask generation step will be inserted into the PIV tutorials and the validation step will be slightly isolated for the purpose of this research. This is to isolate and show the difference or indifference of the analysis results to analyze the validity of using computer vision to generate masks for PIV experiments.

2.3 Zero Shot Learning

Zero shot learning refer to the attempt to classify instances to classes with no previously labeled instances. Countrary to supervised learning, which requires labeling training data with classes for those classes to be used in classification. While recent advancements in supervised learning provides more and more powerful models for classification, the concept itself sufferes from inherent shotcomings such as the inability to work with rare classifications or cases where obtaining data under a certain class is prohibitively expensive. For these reasons, Zero shot learning has been proposed and since its debut, this method has received

high attentiondue to the uniqueness it have on the concept of deep learning algorithms. This research uses models considered to be in the realms of zero shot learning to recognize the expensiveness of obtainnig labeled data in fluid dynamics research.

2.4 Segment Anything Model (SAM) NEED WORK FIGURE WAITING FOR CHANGE

The Segment Anything Model (SAM) developed by Meta AI is used in this project. The model is built to be capable of zero-shot generalization, where it can recognize different objects from an image without further training for the specific use case. The SAM model takes a photo and points indicating the location of the desired object as input to generate a mask. The model is available both online as a demo and as source code which this research benefits from. Figure 2.1 attached below shows the application of SAM in the online demo to generate a mask on one of the PIV images used for this research. Notice how the model was able to recognize the shape of the airfoil when a hollow point is placed on the airfoil. SAM is used as the main model to create the mask once the location of the input box is identified through the use of other machine learning models.

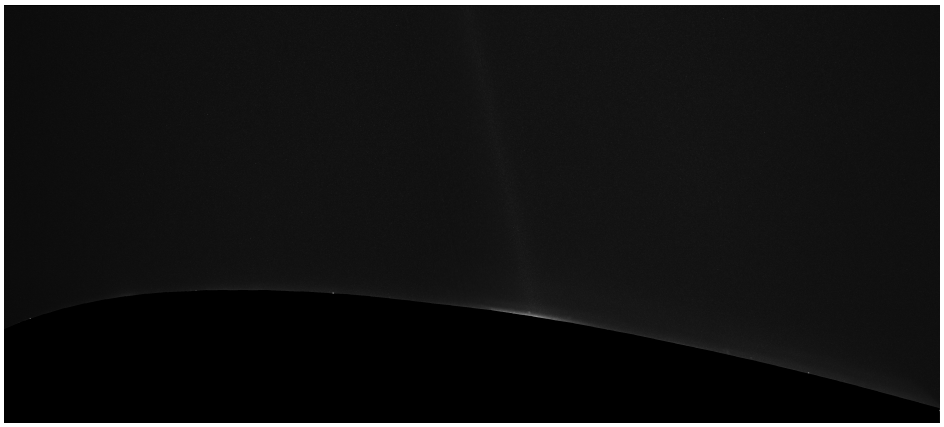

Figure 2.1: PIV Experiment Data: Raw Image (left); Labeled and Masked (right).

Two models are evaluated in this report to study the masking capabilities and efficiency

of the previous generation of SAM and the current generation of SAM. In the first generation of SAM, due to the inability to take natural language input, additional models are employed to supply SAM with input points. The used model is the GroundingDINO model. The later model, SAM3 has the built in capability to take natural language input, therefore the GroundingDINO module is not used with SAM3.

2.5 Grounding Dino

Grounding Dino is the

2.6 Masking by Polynomial Fitting

2.7 Flow Segmentation Using SAM

3 Methodology

Describe how the work was conducted.

3.1 Programming Environment Setup

3.2 Code Preparation and Changes Made

3.3 Validation

3.3.1 Mask Size

3.3.2 Signal to Noise Ratio

3.3.3 Median Tolerances

4 Results

Present results and findings.

5 Discussion

Interpretation and discussion of results.

6 Conclusions

Summary of findings and conclusions.

References

A Appendix A

Begin each appendix on a separate page and label each as Appendix A, Appendix B, etc.

B Appendix B

Additional material.